

Maxime Nicaise

Applied Mathematics / CentraleSupélec & MVA

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Research interests: Optimal transport, statistical learning, sampling and generative models.

EDUCATION

ENS Paris-Saclay M2 Mathematics, Vision and Learning (MVA)	2026–2027
Final-year master's programme, alongside the CentraleSupélec engineering degree.	
CentraleSupélec Diplôme d'ingénieur (Master's level)	2023–2027
Applied mathematics and data science; final year, following a research and industry gap year.	
Lycée International de Valbonne — Classes préparatoires scientifiques	2021–2023

RESEARCH EXPERIENCE

CentraleSupélec / L2S Neural Network Optimization Project	Sep. 2026
Supervisor: Joseph Gabet. Evaluated continuation methods for training neural networks, using progressive resolution and Gaussian smoothing. Compared generalization abilities and loss-landscapes.	
Inria Sophia Antipolis Research Intern, Optimal Transport	Sep. 2025 – Feb. 2026
Supervisor: Samir M. Perlaza.	
- Designed and implemented a stabilized Sinkhorn algorithm for semi-relaxed entropic optimal transport; proved explicit linear convergence rates and evaluated the method numerically. - Established equivalences between optimal transport problems regularized by different f-divergences through transformations of the transport cost.	
CentraleSupélec / MICS Research Project, Functional Analysis & PDEs	Jan. 2024 – Jun. 2025
Supervisor: Anna Rozanova-Pierrat. Studied the Stokes system and Hodge decompositions on domains with irregular boundaries.	

PUBLICATIONS & PREPRINTS

A Gauge-Fixed Sinkhorn Algorithm for Semi-Relaxed Optimal Transport
M. Nicaise, Y. Bermudez, S. M. Perlaza.
Inria research report, 2026. [Report] [Code]
Equivalence of optimal transport problems to regularization on the family of f-divergences
M. Nicaise, Y. Bermudez, S. M. Perlaza.
IEEE International Symposium on Information Theory (ISIT), 2026. [Paper]
On optimal transport with f-divergence regularization
M. Nicaise, Y. Bermudez, S. M. Perlaza.
Inria research report, 2026. [Report]

INDUSTRY EXPERIENCE

Hypertrade, Bangkok Data Science & ML Intern	Feb. 2026 – Aug. 2026
Developed retail demand-forecasting and recommendation models. Contributed to building an analytics agent combining database queries and predictive models on client data. Built an application automating BigQuery queries and allocating Google Cloud compute resources according to workload requirements.	

TEACHING & SCIENTIFIC SERVICE

Mathematics Oral Examiner Lycée International de Valbonne	Sep. 2025 – Feb. 2026
Weekly oral examinations in CPGE MPI/MPI* and PC.	
Mathematics Teaching Assistant CentraleSupélec	Sep. 2024 – Feb. 2025
Designed and taught weekly two-hour tutorials in probability and PDEs for undergraduate students.	
Reviewer: IEEE International Symposium on Information Theory (ISIT), 2026.	

SKILLS & LANGUAGES

Scientific computing: Python (PyTorch, NumPy, SciPy, POT, scikit-learn, pandas), SQL, $\text{\LaTeX}$.
Additional languages and tools: OCaml, R, C, MATLAB, LightGBM, statsmodels.
Languages: French (native), English (proficient), Spanish (intermediate), Thai (conversational).